XRPL Validator — Daily Engineering Report · 2026-05-02

Period: 2026-05-02 00:00 to 24:00 US Eastern (2026-05-02 04:00 to 2026-05-03 04:00 UTC). Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: data.xrpl.org agreement scores. Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Step 1 receipt records (they begin 2026-08-29); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); xrpl-validator merge records (they begin 2026-07-04); amendment data.

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** no receipt records for this day (they begin 2026-08-29).
- **Changes:** 0 deployments, 3 commits (0 with a recorded merge into main; xrpl-validator merge records begin 2026-07-04).

Service levels

No signing records for this day (they begin 2026-09-19).

External agreement, data.xrpl.org (UTC day 2026-05-02): 0.99109 (191 missed of 21,427). *data.xrpl.org scores UTC days from the validations its crawlers received.*

Native engine conformance

No receipt records for this day (Step 1 records begin 2026-08-29, Steps 2–5 records 2026-10-09).

Live counters are kept only for the current run; none exist for a past day.

Receipt register

No receipt records for this day (Step 1 records begin 2026-08-29).

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

No deployment records before 2026-09-11.

Merged into main

Merge records begin 2026-07-04; this day's commits are listed by commit date below.

Findings fixed

No engine findings merged into main this day.

Other engineering changes

- **fix(ffi): suppress no-op Modifieds from mutation-divergence comparison** (`a43897b`, xrpl-validator) Rippled's meta-build at ApplyStateTable.cpp:155-156 skips Modified items where the new SLE bytes equal the pre-state bytes (a "touched but not changed" no-op). The OUTER apply path (line 47-48) calls to.rawReplace(sle) for all modify items unconditionally, so libxrpl's MutationCollector records every touch — including the no-ops that the network meta filtered out.
- **fix(ffi): use closed-ledger OpenView so meta-build runs** (`4242a8e`, xrpl-validator) ApplyStateTable::apply() at line 127 gates the meta-build branch on `if (!to.open() || isDryRun)`. Inside that branch, threadItem and threadOwners run — these add threading-only mutations (PreviousTxnID bumps on related accounts: NFT issuer for transfer-fee NFTs, trustline counterparty for TrustSet, etc.) to the items_ set.
- **fix(ffi): set header.seq = ledger_seq (not -1) for closed-ledger view** (`0c80324`, xrpl-validator) Yesterday's switch from open_ledger to closed-ledger OpenView mode inverted the meaning of header.seq. Open-ledger ctor does `info_.seq = base.seq + 1`, so passing header.seq = ledger_seq - 1 yielded view.seq() = ledger_seq. Closed-ledger ctor copies info_ as-is — no +1 — so passing header.seq = ledger_seq - 1 yields view.seq() = ledger_seq - 1, off by one.

Independence program

No independence record for UTC day 2026-05-02; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

No amendment records for this day.

Open items

None.

Corrections to earlier reports

None.

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
05-02 11:24	—	xrpl-validator	`a43897b`	fix(ffi): suppress no-op Modifieds from mutation-divergence comparison
05-02 11:40	—	xrpl-validator	`4242a8e`	fix(ffi): use closed-ledger OpenView so meta-build runs
05-02 23:29	—	xrpl-validator	`0c80324`	fix(ffi): set header.seq = ledger_seq (not -1) for closed-ledger view

Times in UTC. A commit appears on the day it was committed to main or merged into it.